ENGINEERING CASE STUDY: GATEWAY REDUNDANCY

Comprehensive HSRP & Layer 2 High Availability Network Deployment

Executive Summary: This comprehensive technical report documents the end-to-end design, implementation, and empirical validation of a highly available enterprise edge network topology using Cisco Packet Tracer. By leveraging the Hot Standby Router Protocol (HSRP), multi-tier switch paths, and dynamic routing infrastructure, this architecture completely eliminates single points of failure at the default gateway layer, guaranteeing seamless sub-second host reachability for production client segments (`10.0.1.0/24`).

1. Lab Topology & Architecture Objectives

The deployed network features an enterprise-grade multi-path architecture optimized for fault isolation and high uptime. The structural objectives of this engineering lab include:

- **First-Hop Redundancy:** Provisioning a virtual gateway IP to serve transparently as the egress point for client subnets.
- **Dynamic Edge Routing:** Employing OSPF to interface between core routing entities (`R3`) and the local distribution layer (`R1` , `R2`).
- **Loop-Free Multi-Pathing:** Organizing a robust Layer 2 hierarchy across four interconnected switches to absorb transport link breaks without triggering loops.

2. Deterministic Traffic Path Analysis

Predictable traffic engineering ensures that bandwidth utilization is optimized during steady-state operations while providing a predefined survival path during network emergencies:

- **Primary Path (Steady-State Operations):** Client endpoints utilize the HSRP Virtual IP. Under standard conditions, traffic scales through the lower switched path: `PC1/PC2 → SW1/SW3 → R1 (Active Gateway) → R3 (Enterprise Core)` .
- **Failover Path (Active Failure State):** Upon an active interface or node drop on the primary leg, traffic dynamically realigns via the upper secondary transport: `PC1/PC2 → SW2/SW4 → R2 (Standby Gateway → Transitioned to Active) → R3` .

3. HSRP Protocol Configuration & Convergence Logic

The failover performance of the first-hop gateway relies strictly on fine-tuned HSRP attributes applied across the local distribution layer interfaces (`Gig0/0`):

- **Priority Mapping:** `R1` is assigned an explicit HSRP priority value of `200` to guarantee election as the primary Active gateway. `R2` retains a priority of `50`, placing it strictly in a Standby posture.
- **Preemption Mechanism:** Preemption is intentionally enabled on `R1`. This configuration permits the primary router to automatically reassert its Active dominance following physical link restoration, ensuring the network gracefully reconverges to its optimal engineering path.
- **Convergence Timers:** The deployment uses a 3-second Hello window alongside a 10-second Dead/Hold threshold, establishing deterministic state checking between peers.

4. Layer 2 Switch Infrastructure & Spanning Tree Protocol (STP)

To establish redundant paths at the access tier, a switch matrix consisting of `SW1` , `SW2` , `SW3` , and `SW4` is deployed. Because redundant physical connections create Layer 2 tracking loops, Spanning Tree Protocol (STP) operates natively to block secondary blockings. This safeguards the network from destructive broadcast storms while leaving the block paths primed for instantaneous link failovers.

5. Security Hardening & Edge Best Practices

To align with enterprise production-grade deployments, the following infrastructure security best practices are integrated into the architecture design:

- **HSRP MD5 Authentication:** Configured to prevent malicious or rogue layer-3 devices from injecting unauthorized high-priority HSRP advertisements and executing MitM (Man-in-the-Middle) gateway hijacking.
- **Edge Access Hardening:** Access switch ports connected directly to client endpoints (`PC1` , `PC2`) are constrained with `switchport port-security` and reinforced via `BPDUGuard` to prevent end-user equipment from mis-originating STP topologies.

6. Configuration Summary Matrix

Infrastructure Entity	Feature / Parameter	Assigned Metric	Design Intention
HSRP Group 1	Virtual Gateway IP	`10.0.1.254`	Unified, transparent gateway address for client configurations.
Router 1 (R1)	HSRP Priority / Preempt	Priority `200` / Enabled	Enforces R1 as preferred primary exit node with auto-recovery capability.
Router 2 (R2)	HSRP Priority / Preempt	Priority `50` / Off	Establishes backup failover status; avoids unnecessary active jitter.
Core Pathing	OSPF Routing	Process ID `1`	Dynamically updates upstream enterprise routing tables upon failover.

7. Empirical Failover Validation & Test Logs

Validation data captured from consecutive network stress testing confirms expected high-availability behavior:

Test Phase A: Steady State Validation

With both peers fully operational, a `show standby brief` command executed on `R1` confirms it holds the `Active` state locally, identifying its standby peer at `10.0.1.252` (`R2`). Client validation via an `arp -a` execution on `PC2` confirms that the virtual IP (`10.0.1.254`) successfully resolves to the standard HSRP Virtual MAC structure: `0000.0c07.ac01`.

Test Phase B: Simulated Primary Node Outage

An administrative shutdown was executed on `R1` interface `Gig0/0` via the CLI. HSRP state monitoring triggered an instantaneous migration sequence:

- `R1` entered an `Init` state as its physical interface dropped to a down status.
- Following hold-timer expiration, `R2` successfully shifted from `Standby → Speak → Active` state.
- Upstream OSPF tables converged dynamically, reporting neighbor loss on the primary path and redirecting core egress down the backup route.
- End-user workloads verified uninterrupted external access by maintaining structural ping responses to external destinations (`8.8.8.8`) with zero MAC table reconfiguration required by the clients.

Test Phase C: Link Restoration & Preemption Check

Upon issuing a `no shutdown` command on `R1`, the primary path was re-introduced. Because preemption was active, `R1` successfully measured its priority metrics (`200 > 50`), forcing `R2` to step down gracefully back into a listening `Standby` state. Traffic safely reconverged to the optimal design layout.

8. Portfolio Inclusion Justification

This deployment establishes clean, demonstrable evidence of production-level network capability. It explicitly validates an engineer's competency in foundational corporate network requirements: the mitigation of single points of failure, deterministic configuration alignment, structured command-line verification techniques, and end-to-end multi-layer network troubleshooting.

End of Engineering Report • Gateway Redundancy Optimization Case Study